Demonteverde Agrivet — Operations Improvements

Review of the live PHP POS (Admin / Cashier / Owner) against how a feed-and-vet counter actually runs: sack vs kilo, expiry, layaway, GCash, farm credit, and end-of-day cash. Grounded in the current New_Demonteverde modules, not a generic POS checklist.

Prepared 3 October 2026 · Roles: Admin, Cashier, Owner · Database: agrivet_db

4	7	3	3
Must-fix (P0)	This quarter (P1)	Later (P2)	User roles in the app

The store will feel these first. Cash will not match the drawer whenever anyone pays GCash. Purchase orders do not restock the shelf. A finished sale cannot be reversed. A reprint of a basket is incomplete. Fix those four before adding new modules.

WHAT THE SYSTEM ALREADY DOES WELL

Shift start/close, FIFO stock out, sack-to-kg auto conversion, Display vs Warehouse, near-expiry on the admin dashboard, cashier-selectable promos, layaway stock reserve on create/cancel, SSS/PhilHealth/Pag-IBIG shares, and Owner read-only reports. Build on these instead of replacing them.

SUMMARY OF GAPS

URGENCY	AREA	GAP	REAL-LIFE PROBLEM
P0	Counter	Sale returns and voids after payment	Void only clears the cart. After checkout there is no stock or money reverse.
P0	Counter	Receipt reprint shows one line	A multi-item sale reprints as a single product row.
P0	Cash	Expected drawer includes GCash/Maya/card	Digital sales inflate expected cash; the count always looks short.
P0	Stock	Receiving a PO does not add inventory	Status flips to Received; the shelf is not restocked.
P1	Cash	One shift per cashier per calendar day	No morning/afternoon handoff; start can reopen a closed day.
P1	Cash	Cash-in / cash-out unused	Petty cash and drawer pull-outs have nowhere to go.
P1	Credit	Layaway collections miss the shift	End-of-day cash will not tie to POS sales.
P1	Credit	No farm credit / charge account	Sales are not tied to a customer; utang stays in a notebook.

URGENCY	AREA	GAP	REAL-LIFE PROBLEM
P1	Stock	No physical count or spoilage write-off	Shrinkage and expired meds cannot hit the books cleanly.
P1	Reports	Profit uses current cost, not batch cost	After a mill price hike, COGS and margin are wrong.
P1	HR	Payroll silently overwrites on page load	Peeking at a cutoff rewrites pay records and P&L.
P2	Counter	No held cart, split tender, or supervisor PIN	Peak-hour park, cash+GCash split, and manager override missing.
P2	Stock	One low-stock number for every SKU	A sack of rice and a 10 ml vial share threshold 10.
P2	Reports	No in-app backup / Z-report	XAMPP or power cut can lose a day's work.

P0 — MUST FIX

1. Sale returns and voids after payment

Area: Counter · Evidence: pos.js posVoidOrder(); process_sale.php has no reverse path

Why it hurts. Void only clears the cart before checkout. After a sale posts, there is no way to reverse stock, money, or the receipt. Wrong product, customer change of mind, or spoiled feed after bagging all require a paper workaround.

What to add. Admin-approved return/void on a sale reference: restore FIFO stock, reverse shift totals by payment method, print a return slip, and keep an audit trail.

2. Receipt reprint shows one line of a multi-item sale

Area: Counter · Evidence: Cashier/Receipts.php WHERE s.id = ?; sales is line-item grain

Why it hurts. A cart is stored as several sales rows sharing sale_reference, but Receipts.php loads a single sales.id. The customer gets an incomplete reprint, which fails disputes, GCash claims, and a proper paper trail.

What to add. Reprint by sale_reference: all lines, store name/address, payment method, change, cashier, and a sequential OR number.

3. Expected drawer cash includes GCash, Maya, and card

Area: Cash · Evidence: POS.php expectedDrawerCash; Shift-Report.php same formula

Why it hurts. Closing expected cash is opening + cash_in + total_sales − cash_out. process_sale.php adds every payment method into total_sales. A busy day with digital payments always looks short, so cashiers get blamed or count cash by notebook.

What to add. Split shift totals into cash vs cashless. Expected cash = opening + cash sales + cash-in − cash-out − cash refunds. Show tender mix on the close screen.

4. Receiving a purchase order does not add inventory

Area: Stock · Evidence: Admin/Purchasing.php set_status UPDATE only; free-text item_name

Why it hurts. Marking a PO Received only flips status. No stock_movements IN, no cost update, no expiry batch. Deliveries from feed mills get recorded twice or forgotten, so POS stock and the storeroom diverge.

What to add. Receive against a product SKU: qty, unit cost, expiry, batch. Post IN movement, update on-hand, optionally land cost into cost_price.

P1 — THIS QUARTER

5. One shift per cashier per calendar day, and start can reopen it

Area: Cash · Evidence: cashier_sessions UNIQUE KEY; POS.php start_day upsert

Why it hurts. Unique (cashier_id, session_date) blocks morning/afternoon handoff. Start shift uses ON DUPLICATE KEY UPDATE, which can reopen a closed day and reset started_at. Overnight or split-shift counters cannot close cleanly.

What to add. Allow multiple open/closed sessions per day. Never reopen a closed shift; start a new one. Record counted vs expected variance.

6. Cash-in / cash-out fields exist but are never used

Area: Cash · Evidence: Schema columns; no INSERT/UPDATE of cash_in or cash_out in app code

Why it hurts. Petty cash, change fund top-up, COD to a supplier from the drawer, and owner pull-outs happen every week. The UI shows cash_in/cash_out as zeros, so the close count cannot match reality even if sales tender is fixed.

What to add. Paid-in / paid-out with reason, amount, and who authorized. Tie optional expenses paid from the drawer to the open shift.

7. Layaway collections never hit the cashier shift

Area: Credit · Evidence: Admin/Layaway.php and Cashier/Layaway.php; no cashier_sessions update

Why it hurts. Down payments and balance collections are stored on layaway_payments only. End-of-day cash is higher than POS sales, so the drawer never ties. Overdue layaways have no due date, reminder list, or forfeit rule.

What to add. Post collections to the open shift by tender. Due date, overdue board, partial-release rules, and forfeit/cancel with stock return (already partly there).

8. No farm credit / charge account on the POS

Area: Credit · Evidence: customers table unused at POS; sales has no customer_id

Why it hurts. Customers exist only as a name list. Sales are not linked to a customer. Agrivet shops routinely sell feeds and meds on account until harvest or payday. Today that lives in a notebook, so collections and disputes are invisible to reports.

What to add. Attach customer on checkout, credit limit, outstanding balance, payment-on-account, and aging (7/15/30 days). Block cashier over-limit without admin PIN.

9. No physical count or spoilage write-off

Area: Stock · Evidence: Dashboard low stock < 10 hardcoded; get_near_expiration_products is display-only

Why it hurts. Rats, torn sacks, expired vaccines, and mis-picks make book stock wrong. Display vs Warehouse transfer exists, FIFO OUT exists, but there is no count sheet, variance, or expiry disposal that hits COGS.

What to add. Cycle count by category (feeds, meds, sundries). Variance IN/OUT with reason. Expired-batch write-off from near-expiry alerts.

10. Profit uses current cost, not the batch that was sold

Area: Reports · Evidence: Profit-Loss.php JOIN inventory i ON i.id = s.product_id

Why it hurts. P&L COGS is inventory.cost_price × qty sold. After a price increase from a mill, old bags sold still cost the new price. Net profit is wrong for restocking decisions.

What to add. Stamp unit cost on each sale/stock OUT from FIFO batch. P&L from that cost. Show margin by category (feeds vs vet vs others).

11. Opening Payroll silently overwrites pay records

Area: HR · Evidence: Admin/Payroll.php saveStmt on page load; late TIME(time_in) >= 08:00:00

Why it hurts. Every visit to Payroll.php upserts payroll_records. Peeking at a cutoff rewrites figures and feeds P&L. Cash advances use SUM(amount) in the period, not remaining_balance, so the same utang can hit twice or skip leftovers. Late is hardcoded PHP 30 at 8:00 AM.

What to add. Draft vs Approve payroll. Deduct from remaining_balance. Configurable late rules. OT, rest day, and holiday flags.

P2 — LATER

12. No held cart, split tender, or supervisor PIN

Area: Counter · Evidence: process_sale.php single payment_method; void is in-memory cart only

Why it hurts. Peak hours: a customer leaves to get cash, another wants part GCash part cash, a cashier needs a manager for a price override. Today the cart is lost on refresh and only one payment method is allowed.

What to add. Park/recall sale. Split payments. Manager PIN for manual discount and below-cost sell.

13. Low-stock alert is one number for every SKU

Area: Stock · Evidence: Dashboard-Admin.php stock_quantity < 10

Why it hurts. A sack of rice and a 10 ml vial should not share threshold 10. Conversion rules have per-product thresholds; the dashboard does not. Reorder from mill/vet distributor is still mental.

What to add. Per-SKU reorder point and preferred supplier. Suggest PO from items below point plus fast movers (dashboard already has 7-day top products).

14. XAMPP outage and no in-app backup

Area: Reports · Evidence: Local agrivet_db; DATABASE_RECOVERY_XAMPP.md is manual only

Why it hurts. Power cuts and MySQL crashes wipe a day's work if there is no dump. POS is online-only to local PHP. A cashier cannot sell if Apache is down.

What to add. Scheduled SQL dump from System Settings, restore runbook, and a simple offline queue later. Daily close that saves a Z-report PDF.

RECOMMENDED BUILD ORDER

Week 1–2

- Tender-aware shift totals and cash-in/out
- Receipt by sale_reference
- Return with stock restore
- Receive PO into inventory (SKU, qty, cost, expiry)

Week 3–4

- Layaway money on the shift
- Customer on the sale
- Draft / approve payroll
- Physical count + expiry write-off
- Then credit accounts if farmers still buy on utang

Highest return first: **cash tender on the shift** plus **receipt by sale reference**.

Grounded in New_Demonteverde schema and pages (sales line items, cashier_sessions, purchase_orders, layaways, payroll_records). Revisit after confirming whether this branch still takes farm credit on paper.